

APPENDIX A: SOURCE CODE

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import os
import psutil
import random
import openai
import pyttsx3
from subprocess import Popen
import datetime
import time
import speech_recognition as sr
import webbrowser

# Set your OpenAI API key (you need to get an API key from OpenAI)
openai.api_key = 'your-openai-api-key'

# Path to anti-malware software (example: Windows Defender or any installed
software)
ANTI_MALWARE_PATH = "C:\\Program Files\\Windows
Defender\\MSASCuiL.exe" # Example for Windows Defender

# Initialize pyttsx3 engine for text-to-speech
engine = pyttsx3.init()

def speak(text):
    engine.say(text)
    engine.runAndWait()

def listen():
    recognizer = sr.Recognizer()
    with sr.Microphone() as source:
        print("Listening...")
        recognizer.adjust_for_ambient_noise(source)
    try:
        audio = recognizer.listen(source, timeout=5)
        query = recognizer.recognize_google(audio).lower()
        print(f"You said: {query}")
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        return query
    except sr.UnknownValueError:
        speak("Sorry, I didn't catch that.")
        return None
    except sr.RequestError:
        speak("There was an issue with the speech recognition service.")
        return None

def access_apps_and_files(command):
    if "open" in command:
        app_name = command.replace("open", "").strip()
        try:
            os.system(f"start {app_name}")
            speak(f"Opening {app_name}")
        except Exception as e:
            speak(f"I couldn't open {app_name}. Error: {e}")
    elif "search file" in command:
        file_name = command.replace("search file", "").strip()
        for root, dirs, files in os.walk("C:\\"):
            if file_name in files:
                file_path = os.path.join(root, file_name)
                print(f"File found: {file_path}")
                os.startfile(file_path)
                break
    elif "open browser" in command:
        speak("Opening your web browser.")
        webbrowser.open("https://www.google.com")

def assistant_loop():
    speak("Hello boss, how can I assist you?")
    while True:
        command = listen()
        if command:
            if "exit" in command or "bye" in command:
                speak("Goodbye boss, have a great day!")
                break

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        elif "hey system" in command:
            greet_user()
        else:
            access_apps_and_files(command)

def greet_user():
    current_time = datetime.datetime.now()
    hour = current_time.hour

    if hour < 12:
        greeting = "Good morning boss!"
    elif 12 <= hour < 18:
        greeting = "Good afternoon boss!"
    else:
        greeting = "Good evening boss!"

    welcome_message = f"Welcome! {greeting} I am your Anti-Malware
Assistant."
    speak(welcome_message)
    print(welcome_message)

def scan_for_malware():
    print("Scanning your system for suspicious files and processes...")
    suspicious_files = []
    system_dirs = [r"C:\\Program Files", r"C:\\Windows\\System32", r"C:\\Users"]

    for dir_path in system_dirs:
        try:
            for foldername, subfolders, filenames in os.walk(dir_path):
                for filename in filenames:
                    if filename.endswith(('.exe', '.dll')) and random.random() < 0.05:
                        suspicious_files.append(os.path.join(foldername, filename))
        except PermissionError:
            continue

    for file in suspicious_files:

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    print(f'{file}')

def open_windows_defender():
    try:
        os.system("start windowsdefender:")
        print("Windows Defender opened successfully.")
        speak("Windows Defender opened successfully. Now activating the
assistant.")
        assistant_loop()
    except Exception as e:
        print(f'Error opening Windows Defender: {e}')

def scan_and_protect():
    greet_user()
    scan_for_malware()
    speak("I will now open Windows Defender for you.")
    print("Opening Windows Defender...")
    open_windows_defender()

if __name__ == "__main__":
    scan_and_protect()

```